Week 1 Reading: Process Prediction

1. Definition of the prediction map

A surrogate predicts outputs from specified inputs over a declared domain. A label is a reference output paired with one input row.

Role	This prediction task
Inputs	T: 300–360 K; τ: 1–10 min; C_Af: 0.8–1.5 mol/L
Outputs	C_A in mol/L; conversion X, dimensionless
Fixed context	Ideal, constant-volume, isothermal CSTR; first-order A → B; fixed rate parameters
Reference	Steady-state material-balance solution

Vary feed concentration within the declared domain while keeping reaction assumptions and units the same across rows.

A temperature name and numerical value need units and an operating range to define the prediction task.

2. Derive the steady-state map

The ideal CSTR is perfectly mixed, isothermal, and constant-volume, with first-order reaction A → B. The balance is accumulation = inlet − outlet − reaction:

$$\frac{dC_A}{dt} = \frac{C_{Af} - C_A}{\tau} - k\left(T\right)C_A$$

At steady state, set the derivative to zero. Multiply by residence time, move the concentration terms to the same side, and factor out outlet concentration. Dividing by the positive factor gives:

$$C_A = \frac{C_{Af}}{1 + k\left(T\right)\tau}, \quad X = 1 - \frac{C_A}{C_{Af}} = \frac{k\left(T\right)\tau}{1 + k\left(T\right)\tau}$$

At 330 K, the rate constant is 0.2/min. At 5 min residence time, the product of rate and residence time is 1, so conversion is 0.5. A 1.2 mol/L feed gives a 0.6 mol/L outlet.

Conversion is independent of feed concentration in this first-order model.

3. Build independent train, validation, and test data

Assume the following rate law, with T in K and k in 1/min:

$$k\left(T\right) = 0.2\exp\left[6000\left(\frac{1}{330} - \frac{1}{T}\right)\right]$$

Sample inputs independently and uniformly inside the declared box; compute each label with the steady-state equations. Seed 42 fixes a reproducible sample.

Split	Rows	Role
Train	400	Fit coefficients
Validation	120	Compare model choices
Test	160	Final held-out evaluation

The provided basis is fixed in advance. If comparing another basis or capacity, use validation and leave test untouched.

4. Scale inputs and fit ten features

$$\tilde{x}_j = 2\frac{x_j - l_j}{u_j - l_j} - 1$$

At [330 K, 5 min, 1.2 mol/L], the scaled input is approximately [0, −0.111111, 0.142857]. The scaling uses declared bounds, not statistics computed from test data.

The feature row contains 1 constant + 3 inputs + 3 squares + 3 pairwise products = 10 features. For 400 train rows, Φ has shape (400, 10), Y (400, 2), and B (10, 2).

$$\min_B \|\Phi B - Y\|_F^2, \quad \hat{y} = \varphi\left(x\right)B$$

`np.linalg.lstsq` fits B. Keep B fixed when predicting new input rows.

5. Prediction accuracy and output consistency

$$\mathrm{RMSE}_j = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\left(\hat{y}_{nj} - y_{nj}\right)^2}$$

For each output, subtract the reference from the prediction, square, average over the test rows, and take the square root. Concentration RMSE is in mol/L; conversion RMSE is dimensionless.

The seed-42 baseline gives concentration RMSE 0.032623 mol/L and conversion RMSE 0.025608. Inspect individual errors as well as the averages.

$$r = \hat{C_A} + C_{\mathrm{Af}}\hat{X} - C_{\mathrm{Af}}$$

For feed 1.2, prediction C_A = 0.65 and X = 0.45 gives r = 0.65 + 1.2 × 0.45 − 1.2 = −0.01 mol/L. The predicted pair is not exactly consistent with the defined conversion relation.

6. Answer checks and readings

Try first: at 330 K, 10 min, and feed 1.2 mol/L, what are C_A and X? How many regression coefficients are fitted for two outputs? Why can a low training error be misleading?

Answer check: $k\tau = 2$, so C_A = 1.2/3 = 0.4 mol/L and X = 2/3. B contains 20 coefficients: ten features for each of two outputs. Training evaluates the same points used for fitting; independent errors can be larger.

Check your calculations in the Week 1 lab. Optional background: affine maps and least-squares problems in Chapters 2 and 4 of Boyd and Vandenberghe, *Convex Optimization* (official book PDF).